

Aim :
To execute and verify the special SQL commands.

Procedure:

- step 1: Start
- step 2: Create the table with its essential attributes
- step 3: Insert attribute values into the table
- step 4: Execute different commands and extract information from the table.
- step 5: Stop.

SQL COMMANDS:

1. COMMAND NAME : COMMIT
COMMAND DESCRIPTION: COMMIT command is used to save the work done
2. COMMAND NAME : POINT
COMMAND DESCRIPTION: SAVE POINT command is used to identify a point in a transaction in which it can be restored during roll back command.
3. COMMAND NAME : ROLL BACK
COMMAND DESCRIPTION: ROLL BACK command is used to restore database to original since last commit
4. COMMAND NAME : MAX
COMMAND DESCRIPTION: MAX command is used to find the maximum among the entities in a particular attribute.

5. COMMAND NAME: MIN

COMMAND DESCRIPTION: MIN command is used to find the minimum among the entities in a particular attribute.

6. COMMAND NAME: COUNT

COMMAND DESCRIPTION: COUNT command is used to count the entire entities in a particular attributes.

7. COMMAND NAME: UNION

COMMAND DESCRIPTION: UNION command is used to compile all distinct rows and display all entities in both rows.

8. COMMAND NAME: SUM

COMMAND DESCRIPTION: SUM command is used to add all the entities with the attribute.

9. COMMAND NAME: UNION ALL

COMMAND DESCRIPTION: UNION ALL command is used to return all entities in both rows.

10. COMMAND NAME: INTERSECT

COMMAND DESCRIPTION: INTERSECT command is used to display only similar entities in both rows.

11. COMMAND NAME: MINUS

COMMAND DESCRIPTION: MINUS command is used to display only the rows that don't match in both queries.

12. COMMAND NAME: AVG

COMMAND DESCRIPTION: AVG command is used to find average of entity in particular attribute.

Name	Null?	Type
EMPLOYEE_NAME	NOT NULL	VARCHAR(10)
EMPLOYEE_NO		NUMBER
DEPT_NO		NUMBER
DEPT_NAME		VARCHAR(20)

WORKERS

EMPLOYEE_NAME	EMPLOYEE_NO	DEPT_NO	DEPT_NAME
Arun	101	456	CSE
Ramya	105	584	CSE
Karthik	107	814	IT

SQL > commit;

commit complete.
1 row deleted

EMPLOYEE_NAME	EMPLOYEE_NO	DEPT_NO	DEPT_NAME
Ramya	105	584	CSE
Karthik	107	814	IT

SQL > roll: back;

Roll back complete

EMPLOYEE_NAME	EMPLOYEE_NO	DEPT_NO	DEPT_NAME
Arun	101	456	CSE
Ramya	105	584	CSE
Karthik	107	814	IT

TYPES OF COMMANDS

TCL (TRANSACTION CONTROL COMMANDS)

COMMIT

SAVE POINT

ROLL BACK

SET OPERATORS

UNION

UNION ALL

INTERSECT

MINUS

ARITHMETIC OPERATORS

MAX

MIN

COUNT

AVG

SUM

SQL COMMANDS

EXECUTION:

CREATION OF TABLE

SQL > create table employee (Employee_name varchar(10),
Employee_no number primary key, dept_no number,
dept_name varchar 2(10));

SQL > create table employee1 (Employee_name varchar 2(10),
Employee_no number primary key, (dept_no number,
dept_name varchar 2(10));

Save point created

2 rows updated Inserted

EMPLOYEE_NAME	EMPLOYEE_NO	DEPT_NO	DEPT_NAME
Ayjun	101	456	CSE
Ramya	105	584	CSE
Karthik	107	814	IT
Abhi	104	657	BME
Janu	102	477	IT

Rollback complete.

EMPLOYEE_NAME	EMPLOYEE_NO	DEPT_NO	DEPT_NAME
Ayjun	101	456	CSE
Ramya	105	584	CSE
Karthik	107	814	IT

SUM(EMPLOYEE_NO)

Address on 3182

AVG(EMPLOYEE_NO)

104.333333

MAX(EMPLOYEE_NO)

107

COUNT(EMPLOYEE_NO)

3

MIN(EMPLOYEE_NO)

101

DESCRIPTION OF TABLE

SQL > desc employee1;

SQL > desc employee;

SELECTION OF TABLE VALUES

SQL > select * from employee;

SQL > select * from employee1;

COMMIT & ROLL BACK COMMAND

SQL > select * from employee;

SQL > commit;

SQL > delete from employee where employee-no = 101;

SQL > select * from employee;

SQL > roll back;

SQL > delete * from employee;

SAVEPOINT & ROLLBACK COMMAND

SQL > delete * from employee;

SQL > savepoint ;

SQL > update employee set dept-no = 584, where
employee-no = 105;

SQL > delete from employee where employee-no = 101;

SQL > delete * from employee;

SQL > roll back

SQL > select * from employee;

Employee

EMPLOYEE_NAME	EMPLOYEE_NO	DEPT_NO	DEPT_NAME
Arijun	101	456	CSE
Ramya	105	584	CSE
Karthik	107	814	IT

Employee 1

EMPLOYEE_NAME	EMPLOYEE_NO	DEPT_NO	DEPT_NAME
Arun	105	869	MECH
Kavitha	107	572	IT
Janu	106	578	CSE

Union : EMP_NO

101
105
106
107

Unionall : EMP_NO

101
105
107
105
107
106

6 rows selected

Intersect: EMP_No

105
107

MINUS: EMP_No

101

ARITHMETIC OPERATORS :

SQL > select * from employee ;

SQL > select sum (employee - no) from employee ;

SQL > select avg (employee - no) from employee ;

SQL > select max (employee - no) from employee ;

SQL > select count (employee - no) from employee ;

SET OPERATORS

SQL > select * from employee ;

SQL > select * from employee ;

SQL > SQL > select employee - no from employee union
select employee - no from employee ;

SQL > select employee - no from employee union all
select employee - no from employee ;

SQL > select employee - no from employee intersect
select employee no from employee ;

SQL > select employee - no from employee minus
select * from employee ;

select employee - no from employee ;

ID	NAME	DEPT NAME	SALARY	YEAR
101	Bowling	CSE	20000	2003
102	Bowling	CSE	20000	2002
103	Praveen	CSE	20000	2000

Signature

Result:

this the program has been executed

and verified successfully.